Exploratory Data Analysis on Titanic Dataset

Data Science Training Program — Week 4 Project Report

Project Deliverable: Final Project Analysis Report	**Environment:** VS Code & Python 3
Dataset Source: Titanic Passenger Data (Kaggle/Drive)	**Core Libraries:** Pandas, Matplotlib, Seaborn

1. Executive Summary

This project presents a comprehensive analysis of the Titanic dataset to evaluate passenger survival probabilities based on demographic attributes and class hierarchy. Using statistical data processing and graphical visualizations, key survival trends were identified. The workflow includes reading the dataset, handling missing values, calculating group survival percentages, and creating comparative visualizations.

2. Dataset Overview & Structure

The Titanic dataset contains detailed records of passengers present on the Titanic during its fatal voyage in 1912. The original dataset contains several columns, out of which the primary variables analyzed in this workflow include:

- **Survived:** Target binary variable (0 = Did not survive, 1 = Survived).
- **Sex:** Gender of the passenger (Male / Female).
- **Pclass:** Ticket class assigned to the passenger (1 = 1st Class, 2 = 2nd Class, 3 = 3rd Class).
- **Age:** Numeric age of the passenger.
- **Fare:** Price paid for the passenger ticket.

3. Data Cleaning & Preprocessing Workflow

Raw dataset files often contain missing values or redundant attributes that require cleaning before visualization. In our Python script (`analyse.py`), the following steps were executed:

1. **Missing Value Imputation:** The `Age` column contained missing entries. These were filled using the median age across the dataset to preserve distribution without introducing skewness.
2. **Feature Dropping:** Unused features such as `Cabin`, `Ticket`, and `Name` were dropped to clean the DataFrame and maintain focus on key predictive parameters.

```
# Data cleaning implementation in Python df["Age"] =
df["Age"].fillna(df["Age"].median()) df = df.drop(columns=["Cabin", "Ticket",
"Name"], errors="ignore")
```

4. Data Exploration & Numerical Metrics

Prior to plotting graphs, group mean calculations were performed in Python to assess exact survival rates across gender and class segments:

Feature Segment	Category / Level	Approximate Survival Percentage
Gender (Sex)	Female	~74.2%
	Male	~18.9%
Passenger Class (Pclass)	1st Class	~62.9%
	2nd Class	~47.3%
	3rd Class	~24.2%

5. Exploratory Visualizations & Analysis

To communicate the quantitative findings clearly, two primary visualizations were generated using Seaborn and Matplotlib in VS Code. Both figures are presented below with detailed breakdown notes.

Figure 1: Survival Count by Gender

The first chart compares the total count of survivors and non-survivors grouped by gender (Male vs. Female):

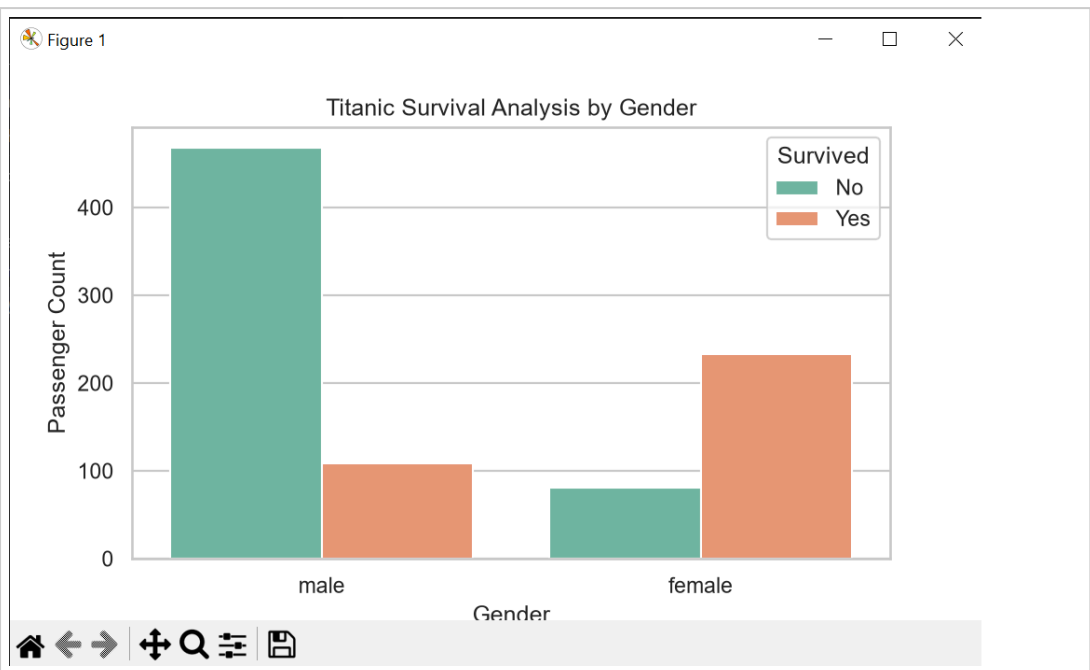


Figure 1: Comparison of survival counts between male and female passengers.

Detailed Analysis of Figure 1:

- **Male Passengers:** Out of nearly 470 male passengers in the dataset, over 460 did not survive (represented by the tall green bar), whereas only around 100 males survived.
- **Female Passengers:** Conversely, for female passengers, over 230 survived (coral bar) out of roughly 310 total females, while fewer than 80 lost their lives.
- **Key Insight:** Gender was the single most dominant factor influencing survival probability on the Titanic. This aligns directly with the "women and children first" maritime protocol followed during emergency life-raft deployment.

Figure 2: Survival Count by Passenger Class

The second chart breaks down survival counts across the three socio-economic passenger tiers (1st, 2nd, and 3rd Class):

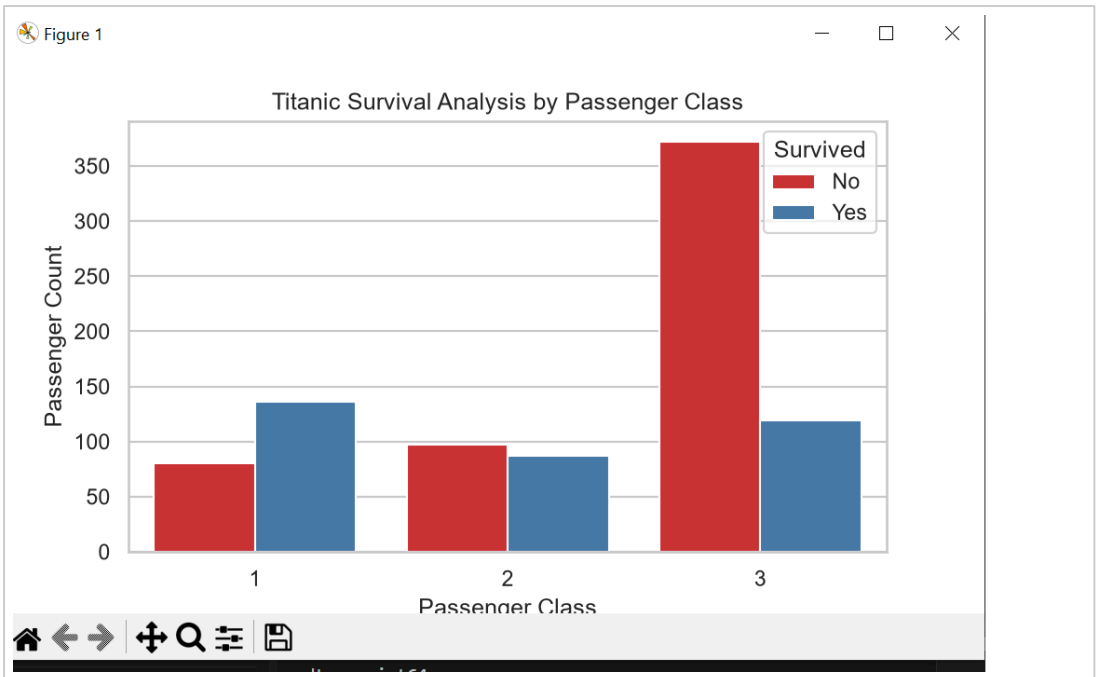


Figure 2: Distribution of survival across 1st, 2nd, and 3rd Passenger Classes.

Detailed Analysis of Figure 2:

- **1st Class:** Passengers in 1st Class experienced the highest survival count (~135 survivors vs. ~80 non-survivors), making it the only class where survivors outnumbered casualties.
- **2nd Class:** Passengers in 2nd Class saw a relatively balanced survival ratio (~90 survivors vs. ~95 non-survivors).
- **3rd Class:** Passengers in 3rd Class suffered the highest casualty rate by a wide margin, with over 370 non-survivors (red bar) and only around 120 survivors (blue bar).
- **Key Insight:** Socio-economic class directly influenced survival odds. First-class passengers had top-deck cabin placement and immediate access to evacuation points, whereas third-class passengers faced structural barriers lower in the ship.

6. Conclusion

> This analysis of the Titanic dataset highlights that female passengers had significantly higher survival rates than males due to evacuation priorities. Additionally, socio-economic status played a decisive role, with first-class passengers experiencing much better survival probabilities than third-class passengers. Data preprocessing steps, such as median age imputation, ensured consistent data quality throughout the analysis. Ultimately, these quantitative patterns align closely with historical accounts of the disaster.